NYAYMITRA 2.0 — Project Report

Problem, solution, beneficiaries, technology, architecture, status, roadmap

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1. Problem statement

Ordinary citizens, particularly in rural India and among those with limited digital or legal literacy, face significant barriers to accessing justice:

- Legal provisions are written in formal, English-heavy language that is difficult to understand.
- Citizens often cannot identify which constitutional article, statute, or government department applies to their situation.
- Free legal aid (NALSA/DLSA, helpline 15100) exists; reported under-utilisation (no supporting usage data provided in this prototype).
- Complaint drafting is intimidating; mistakes can weaken a case or lead to rejection.

2. Why the problem matters

Access to justice is a constitutional goal (Article 39A: equal justice and free legal aid). When citizens cannot understand their rights or reach the right authority, rights remain theoretical. A guided, multilingual, offline-capable navigation tool reduces this gap at low cost.

3. Proposed solution

NYAYMITRA 2.0 is an AI-powered inclusive justice-access and navigation system:

- **Information**: plain-language answers with citations, in English and Gujarati.
- **Navigation**: department directory, triage guides, glossary.
- **Action**: guided complaint drafting with consent, preview, and token-based retrieval.
- **Safety**: no legal verdicts, no outcome guarantees, offline-first by default, PII redacted before local storage.

4. Target beneficiaries

- Rural and semi-urban citizens seeking legal information.
- People with limited English proficiency (Gujarati support).
- Citizens unsure which department or legal-aid body to approach.
- Persons with disabilities (keyboard navigation, focus rings, aria labels — implemented, not yet screen-reader tested).

5. Innovation and uniqueness

- Offline-first deterministic grounding: answers are retrieved from a curated corpus and validated, so the system does not invent legal citations.
- Multilingual (English/Gujarati) with Gujarati digit normalisation for article references.
- Citizen data protection by design: role-based tokens (hashed), rate limiting, redaction before logging, 180-day retention, purge tooling.

• Transparency: every answer carries verification status, warnings, and disclaimers; unverified content is explicitly labelled.

6. Technology used

Backend	FastAPI (Python 3.11), SQLite
Frontend	Vanilla JavaScript, HTML/CSS (no CDN, no inline scripts)
AI / RAG	Retrieval over constitution articles + triage guides; OpenAI-compatible provider optional; offline grounded fallback
Testing	pytest, pytest-cov, jsdom smoke test
Security	pip-audit, CSP, rate limiting, constant-time token comparison
Deployment	Docker, systemd, Render (free-tier shape)

7. System architecture

The backend is a FastAPI application with routers for health, constitution, AI answers, complaints, documents, departments, feedback, and administration. Retrieval uses FTS5 full-text search over articles; triage matches questions to guides; an answer guard validates citations and rejects unsafe legal advice. The frontend is a self-hosted single-page app served from `/web`.

8. Current prototype status

Core features implemented	Verified complete (226 tests pass)
Tests	226 passed, 92% coverage
Database	Migrated + seeded; integrity OK
AI with real model	Blocked (no model runtime)
Legal content verified	Blocked (0 of 9 articles verified)
Accessibility	Implemented, not screen-reader/device tested
Legacy corpus	2,125 quarantined chains; source provenance/copyright unresolved; never served

9. Demonstrable features

• Ask a legal question in English or Gujarati → answer with citations + disclaimer.
• Browse constitution articles and departments.
• File a complaint through preview → consent → confirmation → token receipt.
• Retrieve a saved complaint with reference + token.
• Admin console: stats, verification queue, metrics, backup.

10. Expected social impact (projected, pending field validation)

• Potential for faster, cheaper first-level legal navigation for citizens (impact not yet measured).
• Potential to reduce load on legal aid bodies for simple information queries (pending pilot evidence).
• Greater awareness of rights and available departments (objective, not yet measured).
• A reusable, open prototype for civic technology in India.

11. Future development plan

- Load official Constitution source documents (owner to supply PDFs).
- Complete human legal review of all articles, departments, triage guides.
- Verify AI inference with a live model (Ollama `llama3.2:3b`).
- Conduct real-device, browser, and screen-reader testing.
- Add self-service data deletion and export.
- Pilot with a DLSA or legal-aid NGO.

12. Repository

https://github.com/dhundhwana-arch/Nyaymitra-2.0 Branch: `arena/812d6894-nyaymitra-2-0` | Source snapshot: `cc56d0e` | PR #1 (open, draft)

NYAYMITRA 2.0 — functional software prototype / Proof of Concept — subject to independent evaluation. Not legal advice; not a replacement for courts, lawyers, police, or government authorities. https://github.com/dhundhwana-arch/Nyaymitra-2.0 (branch arena/812d6894-nyaymitra-2-0, commit cc56d0e)